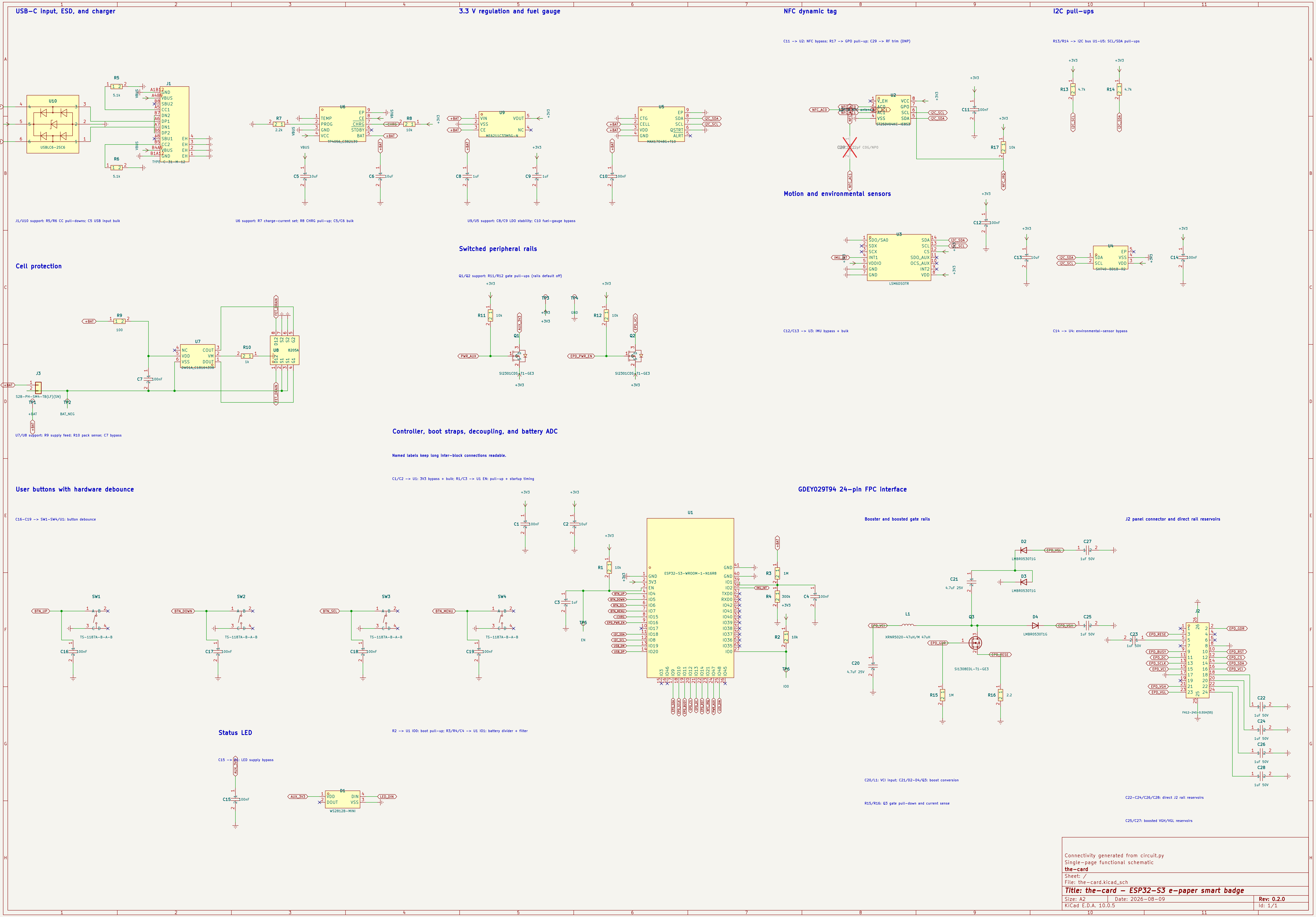
USB-C input, ESD, and charger
J1/U10 support: R5/R6 CC pull-downs; C5 USB input bulk
3.3 V regulation and fuel gauge
U6 support: R7 charge-current set; R8 CHRG pull-up; C5/C6 bulk
U9/U5 support: C8/C9 LDO stability; C10 fuel-gauge bypass
NFC dynamic tag
C11 -> U2: NFC bypass; R17 -> GPO pull-up; C29 -> RF trim (DNP)
I2C pull-ups
R13/R14 -> I2C bus U1-U5: SCL/SDA pull-ups
Motion and environmental sensors
C12/C13 -> U3: IMU bypass + bulk
C14 -> U4: environmental-sensor bypass
Cell protection
U7/U8 support: R9 supply feed; R10 pack sense; C7 bypass
Switched peripheral rails
Q1/Q2 support: R11/R12 gate pull-ups (rails default off)
Controller, boot straps, decoupling, and battery ADC
Named labels keep long inter-block connections readable.
C1/C2 -> U1: 3V3 bypass + bulk; R1/C3 -> U1 EN: pull-up + startup timing
R2 -> U1 IO0: boot pull-up; R3/R4/C4 -> U1 IO1: battery divider + filter
User buttons with hardware debounce
C16-C19 -> SW1-SW4/U1: button debounce
Status LED
C15 -> D1: LED supply bypass
GDEY029T94 24-pin FPC interface
Booster and boosted gate rails
J2 panel connector and direct rail reservoirs
C20/L1: VCI input; C21/D2-D4/Q3: boost conversion
R15/R16: Q3 gate pull-down and current sense
C22-C24/C26/C28: direct J2 rail reservoirs
C25/C27: boosted VGH/VGL reservoirs
Connectivity generated from circuit.py
Single-page functional schematic
the-card
Sheet: /
File: the-card.kicad_sch
Title: the-card — ESP32-S3 e-paper smart badge
Size: A2
Date: 2026-08-09
Rev: 0.2.0
KiCad E.D.A. 10.0.5
Id: 1/1